

Cognitive Kernel v1.7Physics-Aware Governance Kernel Specification

Document Type	Formal Specification (FS) DRAFT-STABLE
Status	Draft-Stable — Implementation Candidate
Version	1.7.0
Supersedes	CK v1.6 — Quantum Adversarial Extension Specification
Date	June 2026
Project	Wolf-CoG-OS / AAIS Federated Cognitive Architecture
Core Additions (v1.7)	GSC (Global State Compiler), IEB v2 (Dual-Channel Internal Event Bus), PTSCR (PhysicsThreat Scope Confinement Rule), STC (Substrate Trust Contract)
Companion Documents	CK v1.5 Governance Superlayer Spec CK v1.6 Quantum Adversarial Extension Spec CIL v1.0 Formal Architecture Spec
Audience	Senior Systems Architects, Distributed Systems Engineers, Governance Engineers, Hardware Abstraction Specialists
Normative Keyword Standard	RFC 2119 — MUST / MUST NOT / SHALL / SHALL NOT / SHOULD / MAY

This document specifies CK v1.7, the direct successor to CK v1.6 (Quantum Adversarial Extension Specification). CK v1.7 is the first version of the Cognitive Kernel to formally elevate physical substrate behavior to a first-class governed input domain — constituting a new architecture class: *physics-aware computational governance infrastructure*. This specification introduces four primary architectural additions: the **Global State Compiler (GSC)**, the **Dual-Channel Internal Event Bus (IEB v2)** with separated Control Plane and Observability Plane channels, the **Substrate Trust Contract (STC)** for Byzantine substrate detection and quarantine, and the **PhysicsThreat Scope Confinement Rule (PTSCR)** for formal authority bounding of the PhysicsThreatAlert subsystem. All CK v1.5 and v1.6 governance precedence hierarchy, invariants, and conformance requirements remain operative and unchanged. This document is normative for CK v1.7 conformance certification.

Table of Contents

- 0.** Architectural Identity — What CK v1.7 Actually Is
 - 0.1 System Classification
 - 0.2 Architectural Maturity Stage
 - 0.3 Canonical One-Paragraph Definition
 - 0.4 Differentiator Statement
- 1.** Version Delta: v1.6 → v1.7
- 2.** Governance Stack (v1.7 — Annotated)
- 3.** The Central Structural Breakthrough
 - 3.1 The Governing Invariant of v1.7

3.2 Formal Substrate Mutability Rule

3.3 Comparison to Prior Architecture Classes

4. Structural Tension Zones (Formally Bounded in v1.7)

4.1 Tension Zone 1 — IEB Monoculture Failure Mode

4.2 Tension Zone 2 — PhysicsThreatAlert Authority Leakage Risk

4.3 Tension Zone 3 — Byzantine Substrate Problem

5. GSC — Global State Compiler

5.1 Problem Statement

5.2 GSC Definition

5.3 SystemStateVector Schema

5.4 Compilation Triggers

5.5 GSC Compilation Guarantees

5.6 GSC as Truth Anchor

5.7 GSC Poisoning Defense

6. Dual-Channel Internal Event Bus (IEB v2)

6.1 Architecture

6.2 Channel Assignment Rules (Canonical)

6.3 Cross-Channel Interaction Rules

6.4 IEB v2 Event Envelope Schema

7. STC — Substrate Trust Contract

7.1 Problem Statement

7.2 STC Honesty Assumptions (Formal)

7.3 STC Verification Protocol

7.4 STC Trust Tier Schema

7.5 STC Trust Tier Definitions

7.6 Byzantine Substrate Response Protocol

7.7 Cross-Substrate Verification

8. PhysicsThreat Scope Confinement Rule (PTSCR)

8.1 Problem Statement

8.2 PTSCR Definition

8.3 PTSCR Enforcement Mechanism

8.4 PTSCR Violation Handling

9. Inter-Layer Coordination Protocol (v1.7)

9.1 Updated Critical Event Chains

9.2 v1.7 Consensus Contract (11 Conditions — Extended)

10. State Vector Schema (Canonical)

10.1 GSC Compilation Rules

11. Failure Mode Registry (Formal Table — Per Layer)

12. Determinism Proof Assumptions

13. Security Threat Model (v1.7 Extended)

14. Conformance Requirements (v1.7)

15. One-Line System Definition

16. Roadmap: CK v1.8

0. Architectural Identity — What CK v1.7 Actually Is

0.1 System Classification

CK v1.7 has crossed a definitive architectural threshold. It is no longer classifiable as a cognitive governance layer in the conventional sense. Through the progressive layering of capabilities across versions v1.0 through v1.7, the Cognitive Kernel has become a merger of four distinct system classes into one coherent architecture. The table below provides the formal system classification map.

System Class	Domain	CK v1.7 Component(s)
Distributed OS Kernel	Classical Computer Science	Epoch Engine, Execution Lifecycle, Internal Event Bus, Scheduling Semantics
Federated Consensus System	Distributed Systems	Consensus Contract v1.7, Quorum + Invariants, Audit Log Immutability
Substrate Abstraction Layer	Systems + Hardware Virtualization	MSL (Meta-Substrate Layer), SAI (Substrate Abstraction Interface), QIGEM / PostLattice
Physical-World Threat Governance System	<i>Novel Class</i>	PhysicsThreatAlert, Decoherence / Lattice Drift / Coherence Collapse Detection, STC , PTSCR

Defining Characteristic of v1.7

The fourth system class — *Physical-World Threat Governance System* — is the defining characteristic of v1.7. CK v1.7 is the first version that treats physical-world substrate behavior as a first-class governed input domain, not an implementation detail. This makes CK v1.7 a **physics-aware computational governance infrastructure** — a class distinct from all prior software

architecture categories.

0.2 Architectural Maturity Stage

The table below maps each version range to its formal architectural maturity designation and characterization.

Version Range	Stage	Meaning
v1.0 – v1.3	Kernel Prototype	Core governance and epoch semantics established
v1.4 – v1.6	Governance System	ASIL, CIEMS, APID, RPDS, FTSS, QIGEM, ZKALS, dAPID layered in; adversarial and quantum extensions applied
v1.7 CURRENT	Substrate-Aware Distributed OS	Physical coupling formalized; substrate promoted to governed variable; GSC, STC, IEB v2, PTSCR introduced
v1.8 (Target)	Deterministic Reconstructable Physics-Governed Compute System	GSC matured to continuous compilation; full cross-layer state reconstruction; hardware Byzantine layer enforced; formal TLA+ proof integration

0.3 Canonical One-Paragraph Definition

CK v1.7 is a substrate-agnostic, event-driven cognitive governance kernel that enforces deterministic execution, physics-aware substrate validation, and federated consensus across heterogeneous execution environments via a formal abstraction layer and immutable event architecture. It is conceptually in the same family as distributed OS kernels (Plan 9 /

exokernel lineage), formal verification systems, fault-tolerant consensus systems (Raft/PBFT class), and quantum-aware control planes — but its differentiating characteristic is that it treats physics as a governed input domain, not merely an execution constraint.

0.4 Differentiator Statement

The following contrast defines the architectural boundary that CK v1.7 crosses relative to all prior versions:

```
// BEFORE CK v1.7 substrate_role          = EXECUTION_ENVIRONMENT
substrate_mutability      = FIXED           // chosen at deployment; immutable
thereafter substrate_observability = INDIRECT // metrics only; no
formal governance surface substrate_governance = NONE //
configuration only; no invariant enforcement // AFTER CK v1.7 substrate_role
= GOVERNED_STATE_VARIABLE substrate_mutability = SELECTABLE //
switchable under GOVERNANCE epoch decision substrate_observability =
FIRST_CLASS // PhysicsThreatAlert, SAI, STC – formal surface
substrate_governance      = FULL           // same invariant + consensus +
audit enforcement as software state
```

This abstraction progression is equivalent in structural depth to the following layered hierarchy:

```
OS abstraction      → hardware abstraction      → physics abstraction layer
(Plan 9 / POSIX)    (hypervisor / HAL)          (CK v1.7 – first formal
instance)
```

No prior architecture in this lineage has formally governed the physics layer as a mutable state variable subject to the same consensus, audit, and invariant enforcement as software state.

1. Version Delta: v1.6 → v1.7

The following table is the normative version delta record. It documents every component-level change between CK v1.6 and CK v1.7, along with the architectural motivation for each change. Components not listed are unchanged from v1.6.

Component	v1.6 Status	v1.7 Change	Motivation
Internal Event	Single-channel; all	Split into	IEB monoculture

Component	v1.6 Status	v1.7 Change	Motivation
Bus (IEB)	event classes co-mingled	CONTROL PLANE CHANNEL (CPC) + OBSERVABILITY PLANE CHANNEL (OPC) — Dual-Channel IEB v2	failure mode — latency coupling, priority inversion, replay ambiguity under mixed-load conditions (see §4.1)
PhysicsThreatAlert	Unlimited escalation authority; no formal scope bounds	PhysicsThreat Scope Confinement Rule (PTSCR) applied — escalation signals only; no direct governance override; IIE-GB mediation required	Hidden second consensus layer risk — PhysicsThreatAlert operated as an informal parallel governance system driven by physical heuristics (see §4.2)
Substrate Trust	Implicit — capability matrix only; no Byzantine detection	Formal Substrate Trust Contract (STC) — honesty assumptions, verification redundancy (BASELINE / CONTINUOUS / DEEP modes), Byzantine detection and quarantine protocol	Byzantine substrate problem — substrate can lie or misreport state; v1.6 had no detection or defense mechanism (see §4.3)
Global State Compiler	Missing — no canonical cross-layer state construct	Introduced as GSC — deterministic compiler from IEB + epoch + substrate → canonical cryptographically-signed SystemStateVector	Replay was partial, audit was fragmented, cross-layer causality was inferred by timestamp correlation rather than enforced by construction (see §5.1)
Governance Stack	8-layer (v1.6); unchanged precedence	8-layer unchanged. New v1.7 components are substrate-plane and infrastructure — they carry no	No governance precedence changes warranted. New components support — not supersede —

Component	v1.6 Status	v1.7 Change	Motivation
		governance precedence position	existing governance hierarchy
Consensus Contract	8 conditions (v1.6)	Extended to 11 conditions : Condition 9 (GSC state vector valid), Condition 10 (STC trust verified for all participants), Condition 11 (PTSCR compliance confirmed)	Close cross-domain coherence gaps opened by new v1.7 substrate and compiler additions
Redundancy Model	7-axis (v1.6)	Extended to 9-axis : + Substrate Redundancy (axis 8) + State Compiler Redundancy (axis 9)	Physics layer and GSC are now formal system components; their failure modes require dedicated redundancy axes
Threat Model	18 threats (T-01 through T-18, v1.6)	Extended to 24 threats : + 6 v1.7-specific threats (T-19 through T-24) covering IEB saturation, channel contamination, Byzantine substrate, GSC poisoning, PTSCR bypass, and cross-substrate Byzantine collusion	Surface and formally bound new attack vectors opened by v1.7 additions

2. Governance Stack (v1.7 — Annotated)

The CK v1.5/v1.6 governance precedence hierarchy is **UNCHANGED** in v1.7. All new v1.7 components operate at the substrate-plane or infrastructure-plane and carry no governance precedence position. The following table provides the complete annotated governance stack with v1.7 status for each layer.

Normative Statement — Precedence Immutability

No v1.7 component **SHALL** be assigned a governance precedence position. The eight-layer governance hierarchy established in CK v1.5 and extended in v1.6 **MUST NOT** be modified by any v1.7 addition, including GSC, IEB v2, STC, and PTSCR. Any proposal to insert a new governance layer **SHALL** require a formal governance amendment epoch, not a CK minor version increment.

Precedence	Layer	v1.6 Status	v1.7 Annotation
1	ASIL — Authorship and Source Identity Layer	Unchanged; lineage proven via ZKALS (v1.6)	Now countersigns GSC SystemStateVector snapshots using ML-DSA-5 (Dilithium-5)
2	IGEM → QIGEM — Quantum-Resistant Identity and Governance Encryption Module	Quantum-resistant key infrastructure (v1.6)	Now encrypts GSC state vectors in transit; provides session key revocation for expelled substrate nodes (STC EXPELLED action)
3	CIEMS — Cognitive Invariant Enforcement and Mediation System	Unchanged; constraint solver + law registry	Now receives GSC -compiled SystemStateVector as primary input to constraint solver; replaces per-layer state

Precedence	Layer	v1.6 Status	v1.7 Annotation
			polling
4	APID → dAPID — Distributed Adversarial Proof Input Defender	Distributed mesh (v1.6)	All dAPID round results now routed exclusively via CONTROL PLANE CHANNEL (CPC) of IEB v2
5	RPDS — Runtime Poisoning Detection System	Unchanged; monitors kernel health	Now monitors GSC compilation health as an additional integrity signal; performs input validation for all GSC compilation inputs
6	FTSS — Federated Trust Scoring System	Unchanged; trust weight management	STC trust scores now feed FTSS as a dedicated substrate-specific trust dimension; STC BYZANTINE status forces FTSS score floor for affected node
7	CONSENSUS ENGINE — Federated Consensus and Quorum	Unchanged; quorum + invariant enforcement	Now requires a GSC -validated SystemStateVector as mandatory consensus input; operates under the extended 11-condition Consensus Contract (v1.7)
8	CK CORE — Cognitive Kernel Core Runtime	Unchanged; execution orchestration	All substrate interactions now governed by STC ; no direct substrate access permitted outside of STC-validated SAI session

New Infrastructure Components (v1.7) — NOT Governance Layers

The following components are introduced in v1.7. They carry **no governance precedence position**. They are substrate-plane and infrastructure-plane components that support, but do not supersede, the governance hierarchy above.

Component	Plane	Role
GSC — Global State Compiler	Cross-cutting / Infrastructure	Deterministic compilation of all layer states into canonical SystemStateVector; truth anchor for replay and audit
IEB v2 — Dual-Channel Internal Event Bus	Infrastructure / Transport	Separated CONTROL PLANE CHANNEL (CPC) and OBSERVABILITY PLANE CHANNEL (OPC); transport substrate for all inter-layer communication
STC — Substrate Trust Contract	Substrate-plane	Formal honesty assumption verification, Byzantine substrate detection, trust scoring, and quarantine protocol for all substrate nodes

Hard Compatibility Rule — LEGACY_V16_MODE

Normative — Backward Compatibility

CK v1.6 nodes connecting to a v1.7 federation **SHALL** operate in LEGACY_V16_MODE. Nodes in LEGACY_V16_MODE **MUST NOT** participate in **GSC** compilation rounds or **STC** verification procedures. The maximum achievable FTSS tier for a node operating in LEGACY_V16_MODE is **PROVISIONAL**. Promotion to TRUSTED tier requires full v1.7 conformance, including GSC participation and STC substrate verification.

3. The Central Structural Breakthrough

3.1 The Governing Invariant of v1.7

The real innovation of CK v1.7 is not the MSL itself, nor any individual component. It is the formal adoption and enforcement of the following single architectural invariant:

"Substrate is no longer a runtime dependency — it is a governed variable."

This invariant has sweeping implications across every aspect of the architecture. The formal before/after state is specified in §0.4. The implications for execution, audit, replay, and security are traced throughout this document.

3.2 Formal Substrate Mutability Rule

A substrate transition — defined as switching the active execution substrate from one registered Substrate Layer (SL) profile to another — is formally classified as a **GOVERNANCE EPOCH**. As such, it is subject to the full governance evaluation chain:

1. Full **dAPID** evaluation of the transition proposal (input-plane adversarial defense)
2. **RPDS** monitoring of the transition execution (runtime integrity)
3. **CIEMS** constraint validation against active law set
4. **ASIL** identity countersignature on the transition authorization
5. **GSC** SystemStateVector recompilation post-transition (substrate_state updated)

6. FTSS trust score update for the new substrate node

Normative — Substrate Transition Rule

Substrate transitions **MUST NOT** occur outside of a GOVERNANCE epoch. Any substrate change that bypasses the Epoch Engine is a **FATAL invariant violation** (INV-2: Layer Separation breach). Detection of an out-of-epoch substrate transition **SHALL** trigger an immediate system HALT and a GovernanceTicket(VIOLATION, INV-2, FATAL) submitted on the CONTROL PLANE CHANNEL.

3.3 Comparison to Prior Architecture Classes

The following table positions CK v1.7 relative to the closest prior architecture classes across four critical capability dimensions.

Architecture Class	Physics Awareness	Substrate Governance	Deterministic Replay	Federated Consensus
Distributed OS (Plan 9 / exokernel)	None	Static — chosen at compile time	Partial — filesystem journal only	None
Fault-Tolerant Consensus (Raft / PBFT)	None	None	Event log only — no state compilation	Full
Formal Verification Systems (TLA+ / Coq)	None	Static — model-fixed	Model-dependent — no runtime	None
Quantum-Aware Control Planes (research systems)	Partial — decoherence detection only	Implicit — no formal contract	None	None
CK v1.7 CURRENT	Full — PhysicsThreat Alert, STC, PTSCR	Full — GOVERNANCE epoch + STC + SAI	Full — GSC canonical SystemStateVector + IEB hash-chain	Full — dAPID mesh + FTSS weighted trust + 11-condition contract

4. Structural Tension Zones — Formally Bounded in v1.7

The following three tension zones were identified in the critical architectural review of CK v1.6. Each represents a structural risk that emerged from the accumulation of capabilities across v1.4 through v1.6 without formal bounding. Each is formally bounded in v1.7 — not deferred.

4.1 Tension Zone 1 — IEB Monoculture Failure Mode

Risk Name: Event Bus Monoculture Failure Mode

Root Cause: The v1.6 Internal Event Bus carries six heterogeneous event classes on a single channel: Consensus events, Epoch transitions, Substrate alerts, Physics threats, Audit logs, and Governance commands. These classes have radically different volume profiles, latency requirements, and priority semantics — and the single-channel architecture cannot enforce separation between them.

Failure Modes:

Failure Mode	Description	Impact
Latency Coupling	High-volume telemetry events delay low-volume governance commands on the shared bus	Governance decisions arrive after the state they govern has changed; INV-4 (Governance Supremacy) is violated in practice
Priority Inversion	A burst of PhysicsThreat events crowds out a pending Consensus event in the queue	Consensus round times out; system incorrectly enters QUARANTINE disposition; false positive governance trigger
Replay Ambiguity	Under high load, interleaved event types on a single channel make causal ordering	Replay produces an incorrect state reconstruction; audit output is non-

Failure Mode	Description	Impact
	ambiguous at replay time	deterministic from the same log
Governance Corruption	Telemetry overload causes governance events to be dropped or delayed beyond their validity window	INV-4 (Governance Supremacy) is effectively violated via infrastructure failure — not adversarial action

Resolution in v1.7: Dual-Channel **IEB v2** — see §6.

4.2 Tension Zone 2 — PhysicsThreatAlert Authority Leakage Risk

Risk Name: Hidden Second Consensus Layer

Root Cause: The v1.6 PhysicsThreatAlert subsystem has, without explicit formal bounds, incrementally acquired four distinct authority types:

7. **Detection authority** — can classify any substrate event as a physics threat
8. **Escalation routing authority** — can trigger governance ticket creation
9. **Human-in-the-loop trigger authority** — can require operator intervention
10. **Runtime pausing capability** — can suspend epoch execution

Without formal scope bounds, PhysicsThreatAlert behaves as a parallel governance system operating alongside Consensus + CIEMS — driven by physical heuristics rather than governed rules. This breaks the clean governance hierarchy established in v1.4 and codified in v1.5.

Governance Hierarchy Violation

PhysicsThreatAlert is **NOT** a governance layer. It has no position in the governance precedence stack. Its authority **MUST** be explicitly and formally limited. Operating without bounds, it functions as a de facto Layer 2.5 — between QIGEM and CIEMS — without the formal constraints that all governance layers carry.

Resolution in v1.7: PhysicsThreat Scope Confinement Rule (PTSCR) — see §8.

4.3 Tension Zone 3 — Byzantine Substrate Problem

Risk Name: Substrate Trust Gap — Substrate Can Lie

Root Cause: The v1.6 SAI (Substrate Abstraction Interface) defines a capability matrix (what a substrate claims it can do) and adapter behavior (how MSL communicates with the substrate). Missing from v1.6 is a formal specification of what happens when a substrate reports false state — whether through failure, misconfiguration, or adversarial intent.

Attack Vectors Opened by This Gap:

Attack Type	Description
False Capability Claim	Substrate reports CAP-QUANTUM but executes classically; determinism contract appears satisfied but quantum-specific guarantees are absent
Metric Fabrication	Substrate reports <code>coherence_window = OK</code> while actually experiencing decoherence drift; PhysicsThreatAlert receives falsely clean input
Silent Failure	Substrate fails internally but reports <code>STATUS: OK</code> to MSL; governance never learns of the failure until post-epoch verification
Adversarial Timing	Substrate artificially delays workload results to manipulate epoch timeout behavior and force a QUARANTINE disposition advantageous to the attacker
State Misreport	Substrate reports <code>pre_state_hash: MATCH</code> when actual hardware state has diverged from the declared pre-epoch state

These constitute the substrate-level equivalent of Byzantine faults in distributed systems theory. No v1.6 mechanism detects, classifies, or defends against this class of failure.

5. GSC — Global State Compiler

5.1 Problem Statement

CK v1.6 contains all the components of a fully auditable, deterministically replayable system — but no single construct that compiles them into a coherent, canonical global state. The consequence of this absence is four distinct operational gaps:

Gap	Effect
Replay is partial	The Epoch Engine replays epoch state; the IEB replays events; but they cannot be jointly replayed to prove a single consistent system state at any given causal point
Audit is fragmented	Each layer maintains its own audit log; cross-layer causality is inferred by timestamp correlation, not enforced by construction — making the audit legally and technically non-authoritative
Cross-layer causality is inferred	A PhysicsThreat that caused an epoch rollback that caused a consensus round failure cannot be <i>proven</i> as a causal chain — only correlated by timestamp proximity
State vector is undefined	There is no canonical answer to "what is the full system state at this moment" that spans all governance layers simultaneously

The **GSC** closes all four gaps with a single deterministic compilation function.

5.2 GSC Definition

The **Global State Compiler (GSC)** is a deterministic, cross-cutting compilation function that takes as input the complete state of all active CK layers and produces a single canonical, cryptographically-signed system state vector.

Formal Definition:

GSC: (IEB_log, EpochState, SubstrateState, GovernanceState, ConsensusState) → SystemStateVector Where: IEB_log = ordered, hash-chained append of all IEB events up to compilation point T EpochState = { current_epoch_id, epoch_status, pre_state_hash, post_state_hash, diff } SubstrateState = { active_substrate_id, sai_profile, stc_trust_score, last_stc_verification } GovernanceState = { open_tickets, active_laws, last_decision, consensus_contract_status } ConsensusState = { quorum_status, participating_nodes, trust_weights, last_consensus_hash } Properties: PURE: No side effects; reads only; does not mutate any layer state DETERMINISTIC: Identical inputs → identical SystemStateVector (guaranteed; see G-1) ISOLATED: Runs in dedicated execution context; no CL/SL logic executes during compilation SIGNED: Output SystemStateVector carries ASIL ML-DSA-5 countersignature

5.3 SystemStateVector Schema

```
{  "state_vector_id":      "SSV-{UUID}",  "compiled_at":
"RFC3339",  "compilation_epoch":  "EPOCH-{UUID}",  "causal_index":
int // monotonically increasing; no gaps; no reuse,  "epoch_state":
{  "current_epoch_id":  "EPOCH-{UUID}",  "epoch_type":
"CREATE | MUTATE | ROLLBACK | GOVERNANCE | CHECKPOINT",  "epoch_status":
"PROPOSED | AUTHORIZED | EXECUTING | COMMITTED | ROLLED_BACK | REJECTED",
"pre_state_hash":      "SHA3-512",  "post_state_hash":  "SHA3-512 |
null" // null if ROLLED_BACK or in-progress,  "governance_context": {
"proposal_id":         "string | null",  "law_context":      ["LAW-
{IDs}"]  },  "substrate_state": {  "active_substrate_id":
"SAI-PROFILE-{ID}",  "substrate_type":      "CLASSICAL | QUANTUM |
LATTICE | HYBRID",  "stc_trust_score":      float // 0.0-1.0,
"stc_verification_timestamp": "RFC3339",  "stc_verification_status":
"TRUSTED | PROVISIONAL | SUSPECT | BYZANTINE",  "active_physics_threats":
["PTA-{UUID}"],  "qigem_epoch":      "EPOCH_0 | EPOCH_1 |
EPOCH_2"  },  "governance_state": {  "open_ticket_count":      int,
"active_law_ids":      ["LAW-{IDs}"],  "last_governance_decision":
"ACCEPT | ROLLBACK | WAIVE | AMEND_LAW | CHANGE_SUBSTRATE | NONE",
"last_decision_timestamp": "RFC3339",  "consensus_contract_status":
"VALID | INVALID | SUSPENDED"  },  "consensus_state":
{  "quorum_status":      "MET | NOT_MET | SUSPENDED",
"participating_node_count": int,  "ftss_weighted_trust_mean":
float,  "last_consensus_hash":      "SHA3-512",
"dapid_mesh_composition_epoch": int  },  "ieb_state":
{  "control_plane_sequence": int // last processed CPC event sequence
number,  "observability_plane_sequence": int // last processed OPC event
sequence number,  "last_event_causal_hash":      "SHA3-512"  },
"integrity": {  "state_vector_hash":      "SHA3-512" // SHA3-
512(epoch_state || substrate_state || governance_state || consensus_state ||
ieb_state || compiled_at || causal_index),  "asil_countersignature":
"ML-DSA-5" // Dilithium-5 signature over state_vector_hash,
```

```
"gsc_instance_id":          "string",      "compilation_duration_ms":
int    } }
```

5.4 Compilation Triggers

The **GSC** compiles a new `SystemStateVector` upon each of the following trigger events. Trigger events are classified by priority and by which channel they arrive on.

Trigger Event	Priority	CPC Source	OPC Source
EPOCH_COMMITTED	HIGH	YES	YES
EPOCH_ROLLED_BACK	HIGH	YES	YES
GOVERNANCE_DECISION	HIGH	YES	NO
CONSENSUS_ROUND_COMPLETE	HIGH	YES	NO
PHYSICS_THREAT_CRITICAL	MEDIUM	NO	YES (via IIE-GB escalation)
STC_STATUS_CHANGE	MEDIUM	NO	YES
PERIODIC_CHECKPOINT	LOW	YES	YES
MANUAL_COMPILE	LOW	YES	YES

5.5 GSC Compilation Guarantees

The **GSC** makes three formal guarantees. These are normative and testable during conformance certification.

G-1 — Determinism: Given identical inputs (`IEB_log`, `EpochState`, `SubstrateState`, `GovernanceState`, `ConsensusState`), the GSC **MUST** produce an identical `SystemStateVector`. The compilation function is pure and side-effect-free. Any GSC implementation that produces non-identical outputs from identical inputs is non-conformant.

G-2 — Completeness: The `SystemStateVector` **MUST** reflect all committed epochs, all governance decisions, and all STC verification results up to compilation point T. No layer state **MAY** be excluded from the compilation inputs. Partial-state `SystemStateVectors` are rejected.

G-3 — Causal Integrity: The `causal_index` is a strictly monotonically increasing integer. No two `SystemStateVectors` **MAY** share a `causal_index` value. The causal chain `SSV[N] → SSV[N+1]` **MUST** be verifiable via `last_event_causal_hash` chaining. Any gap in `causal_index` is a FATAL integrity violation.

5.6 GSC as Truth Anchor

The **GSC** `SystemStateVector` is the canonical truth anchor for four operational functions:

- **Replay:** Replay begins from a **GSC** snapshot (SSV) plus the subsequent IEB log delta. No layer-local state reconstruction is required.
- **Audit Reconstruction:** Any historical system state can be deterministically reconstructed from a prior SSV plus the IEB event delta between that SSV's `causal_index` and the target point.
- **Cross-Layer Causality Proof:** A causal chain such as *PhysicsThreat* → *epoch rollback* → *consensus failure* is provable from two consecutive SSVs without reference to individual layer audit logs.
- **Dispute Resolution:** If two federation nodes disagree on system state, the GSC-compiled SSV bearing a valid ASIL ML-DSA-5 countersignature is authoritative. Countersignature validity is the resolution criterion.

5.7 GSC Poisoning Defense

Because the **GSC** is the truth anchor for all replay, audit, and dispute resolution, it is a high-value attack target. The following defenses are normative:

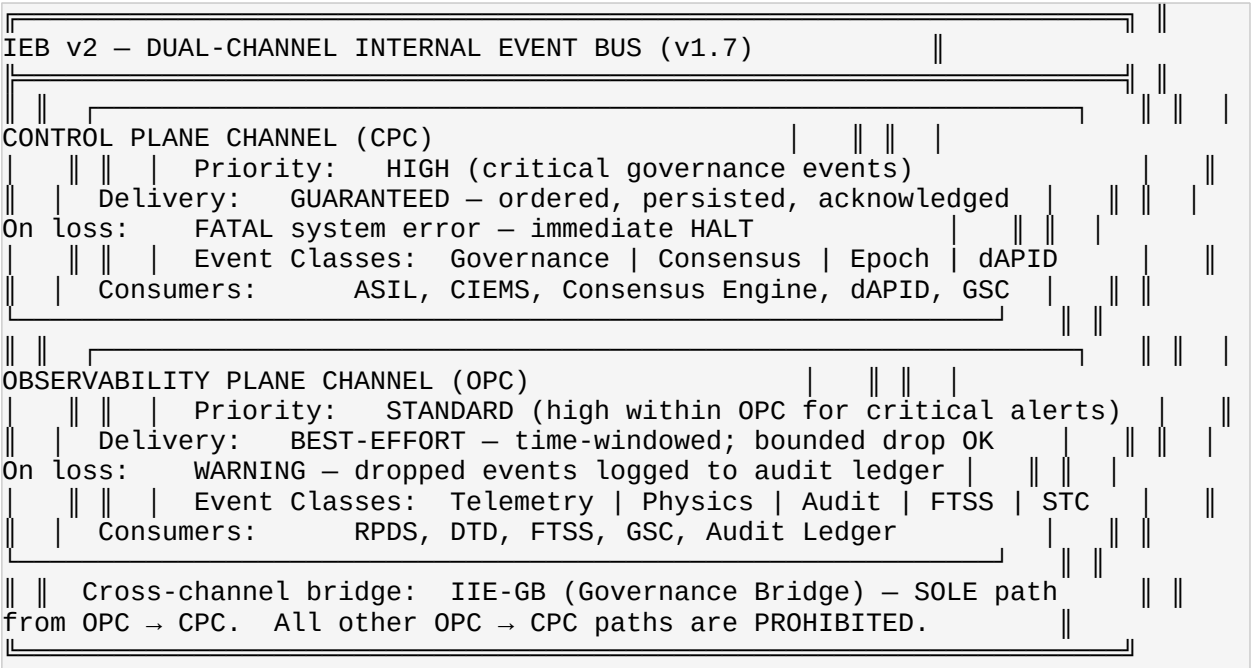
```
{  "gsc_poisoning_defenses": {    "input_validation":      "All inputs to GSC are independently verified by RPDS before compilation begins.",    "output_verification":   "ASIL countersignature (ML-DSA-5) over state_vector_hash is mandatory.",    "compilation_isolation":  "GSC runs in a dedicated isolated execution context.",    "multi_instance_consensus": "In federated deployments, a minimum of 3 GSC instances MUST independently compile the same state and produce matching state_vector_hash values before the SSV is accepted as canonical. Single-instance GSC output is not authoritative.",    "replay_verification":    "Any GSC output SSV input that fails RPDS validation is rejected; compilation is aborted.",    "any_ssv_without_valid_asil_countersignature_must_be_rejected_by_all_consumers": ""  }}
```

MUST be independently verifiable by replaying the IEB log" "from the
prior SSV checkpoint. Verification is deterministic by G-1." } }

6. Dual-Channel Internal Event Bus (IEB v2)

6.1 Architecture

IEB v2 replaces the single-channel v1.6 Internal Event Bus with two strictly separated channels. No event type **MAY** appear on both channels. No event **MAY** be moved between channels after emission. Channel assignment is determined at event type registration time — not at runtime.



6.2 Channel Assignment Rules (Canonical)

Control Plane Channel (CPC) — Assigned Event Types

Event Type	Source Layer	CPC Priority
EpochProposed	CL / IIE-PEV	CRITICAL

Event Type	Source Layer	CPC Priority
EpochAuthorized	IIE-PEV	CRITICAL
EpochCommitted	IIE-PXV	CRITICAL
EpochRolledBack	IIE-PXV / RPDS	CRITICAL
GovernanceTicketSubmitted	IIE-GB	CRITICAL
GovernanceDecisionEmitted	Governance	CRITICAL
GovernanceDecisionAcknowledged	IIE-GB	HIGH
dAPIDRoundResult	dAPID Mesh	HIGH
ConsensusRoundComplete	Consensus Engine	HIGH
ASILEscalation	ASIL	CRITICAL
CIEMSConstraintViolation	CIEMS	HIGH
GSCCompilationComplete	GSC	HIGH
STCStatusChange (governance action required)	STC → IIE-GB	HIGH

Observability Plane Channel (OPC) — Assigned Event Types

Event Type	Source Layer	OPC Priority
PhysicsThreatAlert	DTD / MSL	HIGH (within OPC)
SubstrateMetricUpdate	MSL.StreamMetrics	STANDARD
FTSSTrustDelta	FTSS	STANDARD
RPDSStateHealthReport	RPDS	STANDARD
AuditLogEntry	IIE / All layers	STANDARD
ZKALSProofResult	ZKALS	STANDARD
QIGEMQuantumThreatAlert	QIGEM	HIGH (within OPC)
MeshRecompositionEvent	dAPID	STANDARD
STCVerificationReport	STC	STANDARD
STCStatusChange (observability only)	STC	HIGH (within OPC)

6.3 Cross-Channel Interaction Rules

The following rules are normative and enforced by **IEB v2** channel enforcement logic:

11. OPC events **MUST NOT** trigger direct CPC events without IIE-GB mediation. A PhysicsThreatAlert on OPC that requires a governance action **MUST** be processed by IIE-GB, which emits a GovernanceTicketSubmitted event on CPC.
12. CPC events have delivery guarantees: ordered, persisted, acknowledged. CPC event loss is a FATAL system error requiring immediate HALT.
13. OPC events have time-windowed best-effort delivery. A bounded window of OPC events **MAY** be dropped under extreme load without constituting a FATAL system error. Dropped OPC events **MUST** be logged as AuditLogEntry(OPC_DROP) in the audit ledger.
14. The **GSC** consumes from both channels. The two channel sequence numbers (control_plane_sequence, observability_plane_sequence) are tracked independently in every SystemStateVector.
15. IIE-GB is the sole authorized bridge between OPC and CPC. No other component **SHALL** emit events on CPC in response to OPC events without IIE-GB mediation.

6.4 IEB v2 Event Envelope Schema

```
{  "event_id":          "UUID",    "channel":          "CONTROL_PLANE |  
OBSERVABILITY_PLANE",  "event_type":        "string" // from canonical type  
registry (§6.2),      "source_layer":      "ASIL | IGEM | CIEMS | APID | RPDS |  
FTSS | CONSENSUS | CK_CORE | GSC | STC | GOVERNANCE | IIE_GB",  
"target_layer":        "same enum | BROADCAST",  "sequence_number":  int //  
per-channel; monotonically increasing; no gaps,  "causal_index":      int //  
GSC causal_index at time of emission,  "payload":  
"encrypted_bytes" // QIGEM encrypted for CPC; plaintext permitted for OPC  
audit entries,  "timestamp":          "RFC3339",  "replay_hash":  
"SHA3-512" // SHA3-512(sequence_number || causal_index || payload_hash),  
"prior_event_hash":    "SHA3-512" // hash-chain link to prior event on same  
channel; null for first event }
```

7. STC — Substrate Trust Contract

7.1 Problem Statement

The SAI defines *what* a substrate claims to be able to do. The **STC** defines *what* a substrate must prove about its behavior, and *what* happens when proof fails.

The **STC** formalizes the Byzantine Substrate Problem: a substrate is a Byzantine actor if it deviates arbitrarily from its declared capabilities, reported metrics, or determinism contract. Byzantine deviations may be the result of hardware failure, environmental interference, misconfiguration, or adversarial action. The STC does not distinguish between these causes — it classifies and responds to the observed behavior regardless of root cause.

7.2 STC Honesty Assumptions (Formal)

Every substrate **MUST** satisfy the following six baseline honesty assumptions to be granted TRUSTED status. These assumptions are verifiable and continuously monitored.

```
STC_HONESTY_ASSUMPTIONS {
  HA-1 METRIC_FIDELITY:    Reported metrics
  reflect actual substrate state within declared tolerance T_metric.
  Substrate MUST NOT fabricate, filter, or delay metric reports.
  HA-2
  CAPABILITY_TRUTH:       Declared capabilities are executable.    Substrate
  MUST NOT report capabilities it cannot perform (no phantom capabilities).
  HA-3 DETERMINISM_CONTRACT: Given identical EpochWorkload W and identical
  substrate state S,      execute(W, S) produces identical output_hash on every
  execution.              Substrate MUST NOT produce non-deterministic outputs for
  deterministic workloads.
  HA-4 FAILURE_HONESTY:    Substrate reports
  failures honestly.      Substrate MUST NOT mask, delay, or suppress internal
  failure states.
  HA-5 TIMING_CONTRACT:    Workload execution completes
  within declared timeout bounds T_exec.    Substrate MUST NOT artificially
  delay results beyond T_exec.
  HA-6 STATE_TRUTHFULNESS: pre_state_hash
  reports reflect actual substrate state before epoch execution.    Substrate
  MUST NOT report a pre_state_hash that does not match its actual state. }
```

7.3 STC Verification Protocol

Verification of the honesty assumptions operates in three modes, activated under different conditions:

MODE_BASELINE: Executed at federation admission and after any EXPEL + re-admission event. Full verification of all six honesty assumptions via challenge-response protocol. The substrate **MUST** achieve a composite STC score ≥ 0.90 to

advance from PROVISIONAL to TRUSTED status. Duration: full evaluation cycle required.

MODE_CONTINUOUS: Lightweight real-time monitoring. Runs permanently after MODE_BASELINE completion. Spot-checks HA-3 (determinism contract) and HA-4 (failure honesty) via periodic workload replay with known expected outputs. Monitors HA-1 (metric fidelity) via statistical deviation analysis on reported metrics. No substrate suspension required.

MODE_DEEP: Triggered by RPDS anomaly detection, DTD alert, or a suspicious FTSS score drop for the substrate node. Full re-verification of all six honesty assumptions. The substrate **SHALL** be suspended during MODE_DEEP execution — no EpochWorkloads are accepted during deep verification. If MODE_DEEP reveals no honesty violations, STC status reverts to its prior tier and suspension is lifted.

7.4 STC Trust Tier Schema

```
{  "stc_record": {    "substrate_id":      "SAI-PROFILE-{ID}",    "substrate_type":    "CLASSICAL | QUANTUM | LATTICE | HYBRID",    "stc_trust_status":  "TRUSTED | PROVISIONAL | SUSPECT | BYZANTINE | SUSPENDED | EXPELLED",    "honesty_scores": {      "ha1_metric_fidelity": float // 0.0-1.0,      "ha2_capability_truth": float // 0.0-1.0,      "ha3_determinism_contract": float // 0.0-1.0,      "ha4_failure_honesty": float // 0.0-1.0,      "ha5_timing_contract": float // 0.0-1.0,      "ha6_state_truthfulness": float // 0.0-1.0    },    "composite_stc_score": float // weighted mean of honesty_scores; weights configurable per deployment,    "verification_mode": "BASELINE | CONTINUOUS | DEEP",    "last_verification_timestamp": "RFC3339",    "consecutive_failures": int,    "byzantine_evidence": ["STC-EVIDENCE-{UUID}"],    "remediation_action": "NONE | QUARANTINE | ISOLATE | EXPEL"  } }
```

7.5 STC Trust Tier Definitions

STC Status	Composite Score Range	Meaning	System Response
TRUSTED	0.90 - 1.00	All honesty assumptions met consistently over the observation window	Full execution rights; no restrictions; may participate in Consensus rounds
PROVISIONAL	0.70 - 0.89	Minor deviations detected; within acceptable tolerance	EpochWorkloads accepted; enhanced MODE_CONTINUOUS monitoring;

STC Status	Composite Score Range	Meaning	System Response
			FTSS tier capped at PROVISIONAL
SUSPECT	0.50 – 0.69	Repeated deviations approaching the Byzantine classification threshold	EpochWorkloads held pending STC MODE_DEEP verification; RPDS alerted; GSC flags substrate_state.stc_verification_status = SUSPECT
BYZANTINE	0.25 – 0.49	Systematic, repeated deviation consistent with adversarial behavior or catastrophic failure	Substrate suspended; GOVERNANCE epoch required to resume; ASIL notified; all pending Consensus merges involving this substrate rejected
SUSPENDED	N/A (governance-ordered)	Explicitly suspended by governance decision pending review	No EpochWorkloads accepted; under formal governance review; score collection paused
EXPELLED	0.00 – 0.24	Unrecoverable trust failure; Byzantine behavior confirmed over multiple verification cycles	Substrate ejected from SAI registry; ASIL seals all epoch outputs produced during SUSPECT/BYZANTINE window; QIGEM revokes session keys; re-admission requires full MODE_BASELINE from scratch

7.6 Byzantine Substrate Response Protocol

When `stc_trust_status` = `BYZANTINE`, the following response protocol **MUST** execute in the specified order:

```

ByzantineSubstrateResponseProtocol {
  Step 1: STC emits
  STCStatusChange(status=BYZANTINE, substrate_id, evidence[]) →
  emitted on OBSERVABILITY PLANE CHANNEL (OPC)
  Step 2: IIE-GB monitors OPC;
  receives STCStatusChange(BYZANTINE) → emits
  GovernanceTicketSubmitted(type=COMPOSITE, severity=CRITICAL, source=STC)
  → emitted on CONTROL PLANE CHANNEL (CPC)
  Step 3: RPDS activates MODE_DEEP
  for all kernels that have executed workloads on the affected
  substrate within the SUSPECT observation window
  Step 4: GSC recompiles
  SystemStateVector with:
  substrate_state.stc_verification_status =
  BYZANTINE
  Step 5: Consensus Engine rejects all pending merges involving
  outputs produced by the Byzantine substrate (output_hash provenance
  check)
  Step 6: ASIL seals lineage for all AuthorRecords produced while
  substrate was in SUSPECT or BYZANTINE status (immutable SEALED
  annotation)
  Step 7: Governance Board reviews ticket; issues decision:
  CHANGE_SUBSTRATE (transition to alternate SL) or EXPEL
  Step 8: If decision
  = EXPEL:
  - SAI registry: substrate_id removed
  - QIGEM:
  session keys for expelled substrate node revoked
  - Audit ledger:
  EXPULSION_RECORD created (permanent)
  Step 9: IIE calls AcknowledgeDecision
  GovernanceTicket status → CLOSED
  GSC recompiles final post-
  resolution SystemStateVector }

```

7.7 Cross-Substrate Verification (Multi-Substrate Determinism Check)

For hybrid deployments operating multiple active substrates simultaneously, the **STC** enables cross-substrate verification as an additional Byzantine detection mechanism:

```

{
  "cross_substrate_check": {
    "workload_id": "WL-{UUID}",
    "substrates_compared": ["SAI-A", "SAI-B"],
    "expected_output_hash": "SHA3-512" // derived from prior trusted execution or formal specification,
    "substrate_a_output_hash": "SHA3-512",
    "substrate_b_output_hash": "SHA3-512",
    "hashes_match": bool,
    "deviation_detected": bool,
    "suspect_substrate": "SAI-A | SAI-B | NONE | INDETERMINATE",
    "stc_action": "NONE | ESCALATE_SUSPECT | DEEP_VERIFY_BOTH" } }

```

Resolution Logic:

- If `hashes_match = false` and one hash matches `expected_output_hash`: the deviating substrate is escalated to SUSPECT; STC MODE_DEEP is triggered for that substrate.
- If `hashes_match = false` and both hashes deviate from `expected_output_hash`: STC escalates both substrates to SUSPECT and triggers STC MODE_DEEP on both. The workload output is quarantined pending verification.
- If `hashes_match = true` but both deviate from `expected_output_hash`: this constitutes a Cross-Substrate Byzantine Collusion event (Threat T-24); both substrates are escalated to SUSPECT; full Byzantine investigation is triggered.

8. PhysicsThreat Scope Confinement Rule (PTSCR)

8.1 Problem Statement

PhysicsThreatAlert in v1.6 has accumulated four implicit authority types that, without formal scope bounds, constitute a parallel governance system operating outside the formal governance hierarchy:

- 16. **Detection authority** — can classify any substrate event as a physics threat
- 17. **Escalation routing authority** — can trigger governance ticket creation
- 18. **Human-in-the-loop trigger authority** — can require operator intervention
- 19. **Runtime pausing capability** — can suspend epoch execution

These are appropriate capabilities. The problem is not their existence — it is their lack of formal bounds. Without explicit confinement, PhysicsThreatAlert behaves as a de facto Layer 2.5 governance system driven by physical heuristics rather than governed law — breaking the clean governance hierarchy established in v1.4. This is structurally equivalent to having a second consensus mechanism that operates without the formal invariant enforcement that all first-class governance layers carry.

8.2 PTSCR Definition

The **PhysicsThreat Scope Confinement Rule (PTSCR)** is a hard architectural constraint on PhysicsThreatAlert authority. It is enforced by construction via **IEB v2** channel separation — not by policy configuration.

```
PTSCR: PhysicsThreat Scope Confinement Rule {    PERMITTED
(PhysicsThreatAlert subsystem MAY perform these actions):    - Detect and
classify substrate anomalies (decoherence, lattice drift, coherence collapse)
- Emit PhysicsThreatAlert events to OBSERVABILITY PLANE CHANNEL (OPC) ONLY
- Include a recommended escalation action as advisory payload (non-binding)
- Request STC MODE_DEEP verification (advisory request to STC – not an order)
PROHIBITED (PhysicsThreatAlert subsystem MUST NOT perform these actions):
- Direct override of Consensus Engine decisions    - Direct override of
CIEMS constraint solver decisions    - Unilateral epoch suspension without
IIE-GB mediation    - Direct emission of events on CONTROL PLANE CHANNEL
(CPC)    - Unilateral operator notification without IIE-GB routing    -
```

Direct modification of FTSS trust scores	- Direct invocation of ASIL
lineage sealing	MEDIATION REQUIREMENT: Any PhysicsThreatAlert that
requires governance action, epoch suspension,	operator notification, or
Consensus Engine interaction MUST be routed	through IIE-GB (Governance
Bridge). IIE-GB is the sole authorized mediator	between the OBSERVABILITY
PLANE and the CONTROL PLANE.	IIE-GB evaluates the PhysicsThreatAlert
advisory payload and independently	determines whether a governance-plane
response is warranted. The advisory	payload is informational – IIE-GB is
NOT bound by it. }	

8.3 PTSCR Enforcement Mechanism

IEB v2 enforces **PTSCR** by architectural construction, not by runtime policy check:

PhysicsThreatAlerts are emitted exclusively to the **Observability Plane Channel (OPC)** by type-registration. OPC events cannot directly trigger **Control Plane Channel (CPC)** events — the IEB v2 channel architecture has no direct OPC→CPC event path. The only path from OPC to CPC is through IIE-GB, which applies governance supremacy rules independently before any CPC event is emitted.

This means: a PhysicsThreatAlert can never directly halt an epoch, override a consensus decision, or send an operator notification. The only action available to PhysicsThreatAlert is to signal IIE-GB via OPC, and IIE-GB decides — independently and under full governance rules — whether the situation warrants a governance-plane response.

8.4 PTSCR Violation Handling

If any component attempts to emit a PhysicsThreatAlert-class event directly to the CPC (bypassing IIE-GB), the following response protocol applies:

20. IEB v2 **MUST** reject the emission due to channel mismatch (event type not registered on CPC)
21. Rejection is logged as a `Violation(source=IEB_CHANNEL_ENFORCEMENT, severity=ERROR, event_type=attempted)` entry on OPC
22. The source component's FTSS score receives a violation penalty decrement
23. If the attempt is persistent — defined as ≥ 3 rejected attempts within one consensus cycle — severity escalates to FATAL and ASIL is notified via a `ASILEscalation(type=PTSCR_BYPASS_PERSISTENT)` event on CPC (emitted by IIE-GB)

9. Inter-Layer Coordination Protocol (v1.7)

9.1 Updated Critical Event Chains (v1.7 Additions)

The following event chains are new in v1.7 and represent the normative execution paths for each new architectural scenario. They supplement the event chains defined in CK v1.5 and v1.6 specifications (Chains A through H, which remain unchanged).

Chain I — Epoch Commit with GSC Compilation

Chain I: Epoch Commit with GSC Compilation (v1.7 – Normative) Epoch executes (CK Core + active substrate) → PXV verifies execution (post_state_hash computed) → EpochCommitted emitted on CPC → GSC TRIGGERED: reads CPC sequence N + OPC sequence M → GSC compiles SystemStateVector(causal_index++, epoch_state=COMMITTED, ...) → GSC emits GSCCompilationComplete on CPC → ASIL countersigns SystemStateVector with ML-DSA-5 → SSV stored in audit ledger (immutable) → Downstream consumers (CIEMS, Consensus Engine, FTSS) update state from SSV → CIEMS constraint check passes → epoch FINALIZED

Chain J — Byzantine Substrate Detected

Chain J: Byzantine Substrate Detection and Response (v1.7 – Normative) STC MODE_CONTINUOUS: HA-3 verification fails (output_hash mismatch) → consecutive_failures threshold breached → stc_trust_status → BYZANTINE → STC emits STCStatusChange(BYZANTINE) on OPC → IIE-GB monitors OPC; receives STCStatusChange → IIE-GB emits GovernanceTicketSubmitted(COMPOSITE, CRITICAL) on CPC → RPDS activates MODE_DEEP for affected kernels → GSC TRIGGERED: recompiles SSV(substrate_state.stc_verification_status=BYZANTINE) → Consensus Engine: rejects all pending merges involving Byzantine substrate outputs → ASIL seals lineage for all outputs in SUSPECT/BYZANTINE window → Governance Board: reviews ticket → decides CHANGE_SUBSTRATE or EXPEL → IIE executes decision; SAI registry updated → GSC TRIGGERED: compiles final post-resolution SSV → GovernanceTicket → CLOSED

Chain K — PhysicsThreat Escalation (PTSCR-Compliant)

Chain K: PhysicsThreat Escalation under PTSCR (v1.7 – Normative) DTD classifies: PhysicsThreat(severity=CRITICAL, type=DECOHERENCE_COLLAPSE) → DTD emits PhysicsThreatAlert on OPC ONLY (PTSCR enforcement: CPC blocked) → IIE-GB monitors OPC; receives PhysicsThreatAlert → IIE-GB evaluates: does this warrant governance action? YES → IIE-GB emits GovernanceTicketSubmitted(type=PHYSICS_THREAT, severity=CRITICAL) on CPC → Governance Board reviews → GovernanceDecision emitted on CPC (e.g.,

CHANGE_SUBSTRATE or ROLLBACK) → IIE applies decision → GSC TRIGGERED:
recompiles SSV → GovernanceTicket → CLOSED

Chain L — GSC Poisoning Attempt

Chain L: GSC Poisoning Attempt and Lockdown (v1.7 — Normative) Malicious or corrupted input reaches GSC compilation input stream → GSC input validation (RPDS pre-validation gate) FAILS → GSC rejects compilation; emits GSCPoisoningAlert on OPC → IIE-GB escalates: GovernanceTicketSubmitted(type=GSC_POISONING, severity=FATAL) on CPC → ASIL locks last known-good SSV as canonical truth anchor (LOCKED status) → RPDS activates MODE_TRIGGERED for all affected layers → dAPID: flags and investigates source of malicious input → FTSS: score drop for implicated node(s) → Governance Board reviews; determines scope of compromise → GSC resumes from locked known-good SSV after remediation

9.2 v1.7 Consensus Contract — 11 Conditions (Extended)

The original eight conditions from CK v1.6 remain normative and unchanged. Three conditions are added in v1.7. The binary validity rule is unchanged: any single failure among all 11 conditions constitutes invalid consensus. Partial consensus is not recognized.

#	Condition	Source	Version
1	Quorum of FTSS-weighted nodes has reached agreement on the proposed epoch output	Consensus Engine	v1.5
2	All participating nodes hold <code>STC_STATUS ≥ PROVISIONAL</code> (no EXPELLED nodes)	FTSS / STC	v1.5 (amended v1.7)
3	CIEMS constraint solver finds no active invariant violations in the proposed epoch output	CIEMS	v1.5
4	ASIL has verified authorship lineage of the epoch proposal (ZKALS proof valid)	ASIL / ZKALS	v1.5

#	Condition	Source	Version
5	dAPID mesh has returned a ACCEPT verdict for all inputs in this consensus round	dAPID	v1.6
6	RPDS reports no active poisoning detection events for this epoch chain	RPDS	v1.5
7	QIGEM key epoch is EPOCH_1 or higher (classical-only key material not accepted)	QIGEM	v1.6
8	Epoch pre_state_hash and post_state_hash form a valid, non-colliding pair	Epoch Engine	v1.5
9	GSC SystemStateVector has been compiled and carries a valid ASIL ML-DSA-5 countersignature for this consensus round	GSC / ASIL	v1.7 NEW
10	STC trust status for ALL participating substrate nodes is TRUSTED or PROVISIONAL; no BYZANTINE, SUSPENDED, or EXPELLED substrate participates in this consensus round	STC	v1.7 NEW
11	PTSCR compliance confirmed: no PhysicsThreatAI	IEB v2 / IIE-GB	v1.7 NEW

#	Condition	Source	Version
	ert has directly modified Consensus Engine state in this round without IIE-GB mediation; IEB v2 CPC event log shows no unmediated OPC→CPC escalations		

10. State Vector Schema (Canonical)

The SystemStateVector schema is defined in §5.3. This section defines the normative compilation rules that the **GSC** applies when constructing each SystemStateVector from the raw layer inputs.

10.1 GSC Compilation Rules

```
GSC_COMPILATION_RULES {
  RULE-1 EPOCH_PRIMACY:    epoch_state reflects the most recently COMMITTED epoch.
                           ROLLED_BACK epochs do NOT update pre_state_hash or post_state_hash.
                           A ROLLBACK epoch sets epoch_type = ROLLBACK and restores
                           post_state_hash to the checkpoint being restored.
                           In-progress epochs (EXECUTING) are NOT reflected until COMMITTED.
  RULE-2 SUBSTRATE_CURRENCY:
                           substrate_state reflects the current STC-verified status of the active substrate.
                           If a substrate transition GOVERNANCE epoch is in progress,
                           substrate_state reflects the PRIOR substrate until the transition
                           epoch status = COMMITTED. The pending substrate is not reflected in SSV.
  RULE-3 GOVERNANCE_COMPLETENESS:
                           governance_state.open_ticket_count includes ALL tickets in OPEN or UNDER_REVIEW state.
                           A ticket that is DECIDED but not yet ACKNOWLEDGED by IIE is counted as OPEN.
                           Only CLOSED tickets are excluded from the open_ticket_count.
  RULE-4 CONSENSUS_CURRENCY:
                           consensus_state reflects the outcome of the most recently COMPLETED consensus round.
                           An in-progress consensus round does NOT update consensus_state until completion.
                           A round that times out without quorum updates consensus_state.quorum_status = NOT_MET.
  RULE-5 IEB_TOTALITY:    ieb_state.control_plane_sequence = sequence number of the last CPC event
                           processed before this compilation point T.
                           ieb_state.observability_plane_sequence = sequence number of the last OPC
                           event processed before this compilation point T.
                           ALL events up to these sequence numbers MUST be reflected in the SSV.
                           Events after
```

```

these sequence numbers are NOT included.    RULE-6 CAUSAL_ORDERING:
causal_index is incremented by exactly 1 for every new SSV.    SSV[N+1] MUST
reflect all events processed in SSV[N].    No causal_index may be reused. No
gaps in causal_index are permitted.    A gap in causal_index is a FATAL
integrity violation (see §11 Failure Mode Registry).    RULE-7
HASH_CHAIN_INTEGRITY:    integrity.state_vector_hash =    SHA3-
512(epoch_state || substrate_state || governance_state    ||
consensus_state || ieb_state || compiled_at || causal_index)
integrity.asil_countersignature = ML-DSA-5(state_vector_hash)    Any SSV
whose recomputed hash does not match integrity.state_vector_hash    MUST be
rejected by all consumers as INVALID.    An invalid SSV MUST NOT be used as
a truth anchor for any purpose. }

```

11. Failure Mode Registry — Formal Table (Per Layer)

The following table is the normative failure mode registry for CK v1.7. It covers all identified failure modes per layer, including new v1.7 failure modes. This registry is authoritative for conformance testing and incident response.

Layer	Failure Mode	Root Cause	Detection	Severity	Recovery
CL	Out-of-epoch mutation	Ad-hoc state change bypassing Epoch Engine	PEV pre-check detects stateful operation without active epoch context	FATAL	HALT + GovernanceTicket(VIOLATION, INV-3)
CL	Substrate logic leak	CL contains substrate-specific operations (violates INV-2 Layer Separation)	RG boundary monitor detects substrate-class API call from CL context	FATAL	HALT + GovernanceTicket(VIOLATION, INV-2)
MSL	PhysicsEvent not normalized	SL anomaly event passed raw to IIE	IIE input schema validator rejects non-	ERROR	Block event + request re-normalization

Layer	Failure Mode	Root Cause	Detection	Severity	Recovery
		without MSL normalization	normalized event		on from MSL; log to OPC audit
MSL	Determinism contract breach	output_hash $\neq$ expected_hash for a deterministic workload	PXV post-execution hash verification check	FATAL	ROLLBACK epoch + STC honesty score decrement (HA-3)
SL	Silent failure	Substrate fails internally but reports STATUS: OK to MSL	STC HA-4 continuous verification detects failure_honesty score drop	ERROR	STC status $\rightarrow$ SUSPECT + MODE_DEEP triggered
SL	Byzantine metric report	Substrate fabricates metric values (e.g., coherence_window = OK when drifting)	STC cross-substrate comparison detects metric divergence from peer substrates	FATAL	STC status $\rightarrow$ BYZANTINE + Byzantine Response Protocol (§7.6)
SL	Capability misrepresentation	Substrate claims a capability (e.g., CAP-QUANTUM) it cannot execute	STC HA-2 baseline verification challenge-response failure	FATAL	STC status $\rightarrow$ EXPELLED + ASIL lineage seal for all outputs from this substrate
IEB CPC	Event loss	Infrastructure failure causes CPC event to be dropped	Delivery acknowledgment timeout on CPC guaranteed delivery channel	FATAL	System HALT until CPC integrity is confirmed and event is replayed
IEB OPC	Event burst drop	Telemetry volume	OPC drop counter	WARNING	Log dropped

Layer	Failure Mode	Root Cause	Detection	Severity	Recovery
		spike exceeds OPC time-window capacity	exceeds threshold		events as OPC_DROP audit entries; no system halt required
IEB	Channel cross-contamination attempt	Component attempts to emit a PhysicsThread-class event on CPC	IEB v2 channel enforcement — event type not registered on target channel	ERROR (FATAL if repeated)	Reject emission + Violation log; FATAL escalation after ≥ 3 attempts in one cycle
GSC	Compilation poisoning	Malicious input reaches GSC compilation input stream	RPDS pre-compilation input validation failure	FATAL	Reject compilation + lock last known-good SSV + ASIL escalation (Chain L)
GSC	Hash chain break	SSV integrity.state_vector_hash does not match recomputed hash	GSC integrity verifier on all SSV consumers	FATAL	Reject SSV as INVALID + replay from prior valid SSV checkpoint
GSC	Causal index gap	Missing causal_index value detected in SSV sequence	GSC RULE-6 causal ordering checker	FATAL	HALT + audit investigation to identify missing SSV
dAPID	Quorum failure (timeout)	Too many nodes abstain or disconnect before round completion	Round timeout detector	WARNING	Default QUARANTINE disposition applied to proposal
dAPID	Coordinated falsification	$\geq f$ nodes submit identical	Dissent pattern analyzer	FATAL	ESCALATE_TO_ASIL + FTSS audit

Layer	Failure Mode	Root Cause	Detection	Severity	Recovery
		false verdicts in a coordinated attack	detects abnormal verdict uniformity		for all participating nodes in this round
RPDS	Missed poisoning	Internal corruption evades RPDS detection during epoch execution	PXV post-epoch state hash check reveals post_state_hash mismatch	ERROR	ROLLBACK epoch + trigger RPDS MODE_DEEP for this kernel
FTSS	Score manipulation	Coordinated peer nodes inflate each other's trust scores via behavioral collusion	Multi-round behavioral consistency analysis detects statistical anomaly	ERROR	FTSS audit + CIEMS review of all affected law applications
QIGEM	Key compromise (classical)	Classical cryptographic assumptions broken; EPOCH_0 key material compromised	Anomaly pattern detection or external threat intelligence	CRITICAL	Emergency EPOCH_2 promotion + full re-key of all session material
ZKALS	Nonce replay	A previously used ZKALS proof nonce is submitted in a new authorship context	Nonce registry check on proof verification	FATAL	ASIL escalation + FTSS score floor for implicated author node
ASIL	Lineage collapse	Author identity chain has been flattened into a	Author-Aware Replay + ZKALS cross-verification	FATAL	HALT + replay from genesis to reconstruct full authorship

Layer	Failure Mode	Root Cause	Detection	Severity	Recovery
		system-level narrative (CK authorship identity loss)	of authorship chain		lineage
STC	Trust score staleness	STC verification interval exceeded; trust score is outdated	STC timestamp monitor detects last_verification_timestamp age exceeds T_staleness	ERROR	Force STC MODE_CONTINUOUS re-verification before next EpochWorkload acceptance

12. Determinism Proof Assumptions

For CK v1.7 to guarantee deterministic replay — such that replaying the IEB log from any valid GSC SystemStateVector checkpoint produces an identical system state — the following assumptions **MUST** hold. These are not design aspirations; they are formally verifiable claims, each enforced by a named mechanism.

Assumption ID	Name	Formal Statement	Enforced By
DA-1	IEB Totality	The IEB is the complete and total record of all events that caused state changes. No state change occurs outside of an IEB-recorded event.	IEB v2 dual-channel architecture by construction; any un-bussed state change is an INV-3 violation
DA-2	Epoch Determinism	For any EpochWorkload W	STC HA-3 continuously

Assumption ID	Name	Formal Statement	Enforced By
		and SubstrateProfile P, if the substrate satisfies $STC_STATUS = TRUSTED$, then $execute(W, P)$ produces an identical $output_hash$ on every execution.	verifies; $MODE_DEEP$ confirms under investigation; $EXPELLED$ for persistent violations
DA-3	GSC Purity	The GSC compilation function is pure — it has no side effects and produces identical output given identical inputs.	GSC compilation isolation; no CL/SL logic executes during compilation; RPDS validates GSC execution context
DA-4	ASIL Countersignature Integrity	An ML-DSA-5 countersignature over a $SystemStateVector$ $state_vector_hash$ is computationally infeasible to forge without ASIL's private key material.	$QIGEM_EPOCH_1+$ enforces Dilithium-5 (ML-DSA-5); $EPOCH_0$ (classical) key material not accepted for SSV countersignature
DA-5	Governance Supremacy	No execution path exists that produces a CL state change without a corresponding CPC IEB event. All governance decisions are reflected in the IEB CPC event log in causal order.	IEB v2 CPC guaranteed delivery; any CL state change without CPC event is an INV-4 violation; RPDS monitors
DA-6	Causal Index Monotonicity	The GSC $causal_index$ is strictly monotonically increasing. No two $SystemStateVector$	GSC RULE-6; causal index is a system-global atomic counter; gaps are FATAL integrity violations

Assumption ID	Name	Formal Statement	Enforced By
		rs share a causal_index value. No gaps in the causal_index sequence are permitted.	
DA-7	STC Non-Byzantine	Full deterministic replay is only guaranteed for epochs executed while all active substrates held <code>STC_STATUS ≠ BYZANTINE</code> . Epochs executed during BYZANTINE substrate status are flagged UNRELIABLE in the SSV and require human governance review before inclusion in replay chains.	STC trust status tracked in every SSV <code>substrate_state</code> field; UNRELIABLE flag set by GSC RULE-2 when Byzantine status is active

Note — Replay Scope

The seven assumptions above collectively define the scope of CK v1.7 deterministic replay. Full deterministic replay is achievable for any system state within the bounds of all seven assumptions. Epochs that violate any assumption are flagged in the `SystemStateVector` and may require human-governed remediation before inclusion in authoritative replay chains.

13. Security Threat Model (v1.7 Extended)

The v1.6 threat model (Threats T-01 through T-18) remains operative and unchanged. This section documents the six new threats introduced by v1.7 architectural additions. These threats surface attack vectors that did not exist prior to the introduction of IEB v2, GSC, STC, and PTSCR.

Threat #	Threat Name	Attack Vector	Defending Layer(s)	Detection Method	Response Protocol
T-19	IEB CPC Saturation	Adversary floods CPC with forged or replayed governance events, causing genuine governance commands to be delayed beyond their validity window	IEB v2 rate limiter, dAPID source validation, ASIL event signature verification	CPC delivery latency threshold exceeded; event signature verification failure rate spikes	Rate-limit adversarial source + FTSS score floor for implicated nodes + ASILEscalation(CPC_SATURATION)
T-20	IEB Channel Cross-Contamination	Adversary or compromised component emits a PhysicsThreat-class event directly on CPC to gain unmediated governance influence, bypassing IIE-GB	IEB v2 channel type enforcement; event type registry	Channel mismatch: event type not registered on target channel; IEB enforcement log	Reject emission + Violation log (OPC) + FTSS penalty; FATAL escalation after ≥ 3 attempts per cycle
T-21	Byzantine Substrate	A substrate node	STC HA-3 continuous	Deterministic contract	STC status → SUSPECT

Threat #	Threat Name	Attack Vector	Defending Layer(s)	Detection Method	Response Protocol
	(Silent)	silently fails or is adversarially compromised, continuing to report STATUS: OK while producing incorrect or non-deterministic epoch outputs	deterministic verification, STC HA-4 failure honesty, PXV post-epoch hash check	breach detected by PXV; HA-4 failure honesty score drops below threshold	→ MODE_DEEP → BYZANTINE if confirmed; full Byzantine Response Protocol (§7.6)
T-22	GSC State Vector Poisoning	Adversary injects malicious or corrupted events into the GSC compilation input stream, causing the GSC to produce a false SystemStateVector that becomes the canonical truth anchor	RPDS pre-compilation input validation, ASIL ML-DSA-5 countersignature, multi-instance GSC consensus	RPDS input validation failure; multi-instance hash mismatch; SSV hash recomputation failure	Reject SSV + lock last known-good SSV + GovernanceTicket(GSC_POISONING, FATAL) + ASIL escalation (Chain L)
T-23	PTSCR Bypass Attempt	A compromised DTD or MSL component attempts to emit a PhysicsThreatAlert directly on CPC,	IEB v2 channel enforcement; PTSCR type registry lock	Channel mismatch rejection by IEB v2 enforcement layer; rejection logged	Reject + ERROR Violation log; FATAL escalation after ≥3 attempts per consensus cycle; FTSS score

Threat #	Threat Name	Attack Vector	Defending Layer(s)	Detection Method	Response Protocol
		bypassing IIE-GB mediation to gain direct governance influence			decrement for source
T-24	Cross-Substrate Byzantine Collusion	Two or more substrate nodes are coordinated to produce matching but incorrect output_hashes, defeating single-substrate Byzantine detection by making the falsified output appear consistent	STC cross-substrate verification with known expected_output_hash, PXV post-epoch hash check	Both substrate hashes match each other but deviate from expected_output_hash derived from formal specification or prior trusted execution	Both substrates → SUSPECT; STC MODE_DEEP triggered for both; ASIL lineage seal for all affected outputs; GovernanceTicket(CROSS_SUBSTRATE_COLLUSION, FATAL)

14. Conformance Requirements (v1.7)

All v1.6 conformance requirements (C-01 through C-22 from CIL v1.0) remain operative and unchanged. The following requirements are added in v1.7.

Conformance to CK v1.7 requires satisfaction of ALL requirements from C-01 through C-34.

Req. ID	Requirement	Normative Force	Verified By
C-23	IEB v2 is deployed as a dual-channel bus. The CONTROL PLANE CHANNEL and OBSERVABILITY PLANE CHANNEL are strictly separated with no event appearing on both channels.	MUST	IEB channel audit; event type registry inspection
C-24	All CPC events have guaranteed ordered delivery and delivery acknowledgment. CPC event loss MUST be treated as a FATAL system error requiring immediate HALT.	MUST	CPC delivery acknowledgment log; failure injection test
C-25	All PhysicsThreatAlerts MUST be emitted exclusively to OPC. Direct CPC emission by PhysicsThreat subsystems is prohibited.	MUST NOT	IEB type registry inspection; PTSCR channel enforcement log review
C-26	IIE-GB is the sole authorized mediator between OPC and CPC. No other component may emit CPC events in response to OPC events without IIE-GB mediation.	MUST	CPC event source audit; IIE-GB mediation log verification
C-27	GSC MUST be present and MUST compile a new SystemStateVector on every EPOCH_COMMITTED and GOVERNANCE_DE	MUST	GSC compilation trigger log; SSV causal_index continuity verification

Req. ID	Requirement	Normative Force	Verified By
	CISION event. Compilation MUST not be skipped or deferred for these trigger events.		
C-28	All GSC output SystemStateVectors MUST carry a valid ASIL ML-DSA-5 (Dilithium-5) countersignature. Unsigned or classically-signed SSVs are not conformant.	MUST	SSV integrity.asil_countersignature verification on all stored SSVs
C-29	The GSC causal_index MUST be strictly monotonically increasing with no gaps, no repeated values, and no resets outside of a genesis event.	MUST	Causal index sequence audit; gap detection test
C-30	STC MUST be present and MUST actively verify all substrate nodes. STC verification mode MUST be at minimum MODE_CONTINUOUS for all nodes that have completed MODE_BASELINE.	MUST	STC verification_mode field inspection; last_verification_timestamp age check
C-31	No EpochWorkload MUST be accepted from or assigned to a substrate node with stc_trust_status = BYZANTINE or EXPELLED.	MUST NOT	STC trust status gate on EpochWorkload assignment; Byzantine injection test
C-32	Substrate transitions MUST occur exclusively via GOVERNANCE	MUST	Epoch type audit on all substrate transitions; out-of-epoch transition

Req. ID	Requirement	Normative Force	Verified By
	epoch type. Any substrate transition outside of a GOVERNANCE epoch is a FATAL INV-2 violation.		detection test
C-33	The v1.7 Consensus Contract (11 conditions, §9.2) is the operative consensus validity definition. All 11 conditions MUST be satisfied for a consensus round to be valid. No partial consensus is permitted.	MUST	Consensus validation log; condition 9, 10, 11 specific verification tests
C-34	PTSCR MUST be enforced. No PhysicsThreatAlert MAY directly modify Consensus Engine state without IIE-GB mediation. This is enforced by IEB v2 channel separation by construction.	MUST	IEB v2 channel enforcement log; PTSCR bypass injection test; IIE-GB mediation audit

15. One-Line System Definition

CK v1.7 is a physics-aware, substrate-governed, federated cognitive kernel that enforces deterministic execution across heterogeneous substrates via a dual-channel event bus, compiles authoritative cross-layer system state through a Global State Compiler, detects and quarantines Byzantine substrates via a formal Substrate Trust Contract, and confines physical

threat governance to bounded escalation signals — all within the unchanged CK v1.5/v1.6 governance precedence hierarchy under ASIL + CIEMS dual authority.

16. Roadmap: CK v1.8

Architectural Directive — v1.8 Scope

CK v1.8 follows the architectural directive: *"Do NOT expand layers further yet. Formalize runtime behavior, not architecture."* v1.8 targets are exclusively runtime formalization, proof integration, and self-healing capabilities within the existing layer structure. No new governance layers, substrate-plane components, or infrastructure-plane components are proposed for v1.8.

#	v1.8 Target	Description	Foundation in v1.7
1	GSC Maturation — Continuous Compilation	GSC moves from event-triggered compilation to fully autonomous continuous compilation with sub-epoch granularity. The causal chain becomes the primary unit of system reasoning, enabling real-time cross-layer causality proofs without waiting for epoch boundaries.	GSC (§5) — trigger model extended to continuous; causal_index granularity refined
2	Autonomous Trust	FTSS detects and	FTSS + STC trust

#	v1.8 Target	Description	Foundation in v1.7
	Graph Healing	autonomously heals minor trust degradations without CIEMS arbitration for scoring deviations below a defined self-healing threshold. CIEMS arbitration is retained for threshold breaches. This reduces governance overhead for routine trust fluctuations.	score integration (§7.5) — provides the trust data foundation for self-healing logic
3	Cognitive Load Balancing Protocol	CK Core workload distribution across TRUSTED + SOVEREIGN federation nodes is governed by FTSS trust weights and substrate STC composite scores. Load distribution decisions are GOVERNANCE epochs — no ad-hoc load balancing permitted.	STC trust tier definitions (§7.5) + GOVERNANCE epoch for substrate transition (§3.2)
4	Formal Failure Mode Proofs	Formal proof (TLA+ / Coq-compatible specification) of the failure modes in §11 of this document. Machine-verifiable invariant checking as the basis for conformance certification, replacing human-verified conformance testing for select	Determinism Proof Assumptions (§12) provide the formal base; GSC causal chain enables TLA+ state representation

#	v1.8 Target	Description	Foundation in v1.7
		failure modes.	
5	ASIL Temporal Binding	Cryptographic timestamp binding to all AuthorRecords — provable non-repudiation that an AuthorRecord existed at a specific causal_index. Closes the temporal forgery gap: an adversary cannot claim an AuthorRecord was created at a causal_index earlier than it actually was.	GSC causal_index (§5.3) + ASIL ML-DSA-5 countersignature (§5.3) — binding causal_index to authorship timestamp
6	Self-Healing dAPID Mesh	The dAPID mesh automatically detects and quarantines Byzantine nodes mid-round without requiring a full mesh recomposition. Quarantine is a local consensus decision within the mesh — no CIEMS escalation required for intra-round Byzantine node isolation below the threshold requiring governance intervention.	dAPID failure mode coverage (§11) + STC Byzantine detection (§7.5) — mesh applies analogous trust model locally

v1.8 Goal Statement

A system whose runtime behavior is as formally specified as its architecture — where every state, transition, failure, and recovery is provable from first principles given the IEB log and a genesis SystemStateVector, without reference to any layer-local state.

Document	Cognitive Kernel v1.7 — Physics-Aware Governance Kernel Specification
Specification ID	CK-FS-1.7.0
Project	Wolf-CoG-OS / AAIS Federated Cognitive Architecture
Status	Draft-Stable — Implementation Candidate
Date	June 2026
Conformance Baseline	C-01 through C-34 (this document); C-01-C-22 inherited from CIL v1.0
Supersedes	CK v1.6 — Quantum Adversarial Extension Specification
Normative Keyword Standard	RFC 2119
Cryptographic Standards	SHA3-512 (NIST FIPS 202), ML-DSA-5 / Dilithium-5 (NIST FIPS 204)
Next Version Target	CK v1.8 — Deterministic Reconstructable Physics-Governed Compute System

END OF DOCUMENT — CK v1.7 Formal Specification (FS) — CK-FS-1.7.0 — Draft-Stable

Wolf-CoG-OS / AAIS Federated Cognitive Architecture | June 2026